MIL-V-8419A(USAF)
24 February 1964
Superseding
MIL-V-8419(USAF)
31 August 1953

MILITARY SPECIFICATION

VACUUM PUMP, ROTARY, POWER DRIVEN

1, SCOPE

1.1 This specification covers one type of rotary power driven vacuum pumping unit, 0.50 inches of mercury absolute ultimate vacuum, complete with vacuum storage tank, electric motor drive, and automatic controls.

1.2 Classification. - The vacuum pumps shall be of the following minimum sizes, as specified (see 6.2).

Size 1 - 100 CFM free air displacement
Size 2 - 200 CFM free air displacement

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium, Electrodeposited
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fireboard, Wood Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-C-6529	Corrosion-Preventive Compound, Aircraft Engine
MIL-P-7105	Pipe Threads, Aeronautical, National Form, Symbol ANPT
MIL-E-7729	Enamel, Gloss, Aircraft Application

FSC 4310

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MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-T-9107	Test Reports, Preparation of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U S Military Property

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. - This specification provides for preproduction testing.

3.2 Components. - The vacuum pump shall be complete with all components The components shall include, but not necessarily be limited to, the following: vacuum pump, motor, drive, vacuum tank, and all necessary controls and piping for complete operation as a vacuum pumping unit.

3.3 Materials. - All parts of the vacuum pumping unit that are normally subject to corrosion shall be constructed from corrosion resistant materials. Steels shall be protected from corrosion, as necessary (see 3.3.1), by cadmium plating in accordance with Specification QQ-P-416, Type II or III, and of a class that is adequate to achieve the degree of protection required, or other equally suitable means.

3.3.1 Protective treatment. - When materials used in the construction of the pump are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions, shall be avoided.

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3.4 Design. - The vacuum pumping unit shall be designed for 500 hours of operation at rated capacity and pressure between overhauls and to minimize loss of service time for maintenance, repair and periodic servicing.

3.4.1 Type. - The vacuum pump shall be of the rotary oil sealed type either air or water cooled.

3.4.2 Maintainability. - Features providing the most expeditious and easy maintenance of the equipment consistent with performance, reliability, interchangeablity, and costs, shall be to the maximum extent practicable. The use of special tools, not normally available commercially, and the need for special maintenance training shall be kept to a minimum.

3.4.3 Safety. - Safety devices necessary for the protection of operating personnel shall be provided. All moving parts which are of such nature or so located as to be hazardous to operating or maintenance personnel shall be fully enclosed or properly guarded.

3.5 Construction. - The vacuum pumping unit shall be so constructed that no parts will work loose in service. The unit shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation and service.

3.6 Performance. - The vacuum pump shall be capable of meeting the performance requirements specified herein.

3.6.1 Free air pumping capacity. - The free air capacity of the vacuum pump shall be such that when it is operated at rated speed with both inlet and discharge connections open to standard atmospheric pressure, the quantity of air pumped through the unit shall be a minimum of 100 cfm or 200 cfm as specified (see 6.2).

3.6.2 Operating time. - The vacuum pump shall be capable of continuous operation at the design capacity for a period of not less than 12 hours without requiring shutdown for servicing, maintenance or repairs.

3.6.3 Ultimate vacuum. - When operating at rated speed in an ambient temperature of +70 °F with the inlet completely blanked off except for a connection to an absolute pressure measuring device and the discharge open to atmospheric pressure, the vacuum pump shall be capable of attaining and holding an absolute pressure of 0.50 inches of mercury absolute or less.

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3.7 Lubrication. - The vacuum pump shall incorporate an automatic lubrication system. A solenoid valve, when necessary, shall be provided in the oil feed line to control the flow of oil during operation of the pump. The oil system shall be complete with a drain valve, oil level gage, and filling plug. Means shall be provided for removing water vapor from the pump oil.

3.8 Separator. - The pump shall incorporate a separator arranged to effectively remove oil and raw moisture from the exhaust gases. The separator shall be provided with a sump for collection of oil and water. The sump shall be provided with a drain valve.

3.9 Base. - The pump and motor shall be mounted on a common substantial base which shall be constructed to permit the base to rest solidly against any flat surface.

3.10 Motor. - An electric motor, suitable for operations specified herein shall be supplied with each vacuum pump. The motor shall be designed to operate on 220/440 volts, 60 cycle, 3 phase, unless otherwise specified (see 6.2).

3.11 Drive. - The pump shall be connected to the motor by means of a multiple "V" belt drive. Suitable manual or automatic adjustment shall be provided for keeping the "V" belt tight.

3.12 Control panel. - A control panel for wall mounting shall be furnished with each pump. The panel shall be constructed of material of approved commercial quality suitable for this type of equipment.

3.12.1 Starter - The control panel shall incorporate a "start-stop" push button control station which activates an enclosed across-the-line type magnetic motor starter. The motor starter shall be mounted on the motor-pump base (see 3.9). The starter shall be equipped with a Hand-Off-Automatic selector switch to permit either automatic (vacuum switch controlled) or manual operation. Adequate manual reset overload protection shall be provided.

3.12.2 Vacuum regulator. - A vacuum regulator shall be provided on the panel for maintaining a constant vacuum in the storage tank. The vacuum regulator shall have a two inch differential switch to start and stop the vacuum pump at any desired vacuum up to 30 inches of mercury. The vacuum regulator shall be equipped with a bellows actuated absolute pressure gage.

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3.13 Electrical equipment. - All electrical equipment and wiring shall conform to the standards of the American Institute of Electrical Engineers and National Electrical Manufacturers Association.

3.14 Vacuum storage tank. - A vacuum storage tank shall be supplied with each vacuum pump. The vacuum storage tank shall be capable of meeting the vacuum test as specified in 4.6.5 and shall be designed and constructed in accordance with the ASME Unfired Pressure Vessel Code.

3.14.1 Size 1. - The Size 1 vacuum pumping unit shall be complete with a vertical storage tank approximately 2 feet in diameter by 8 feet high and a minimum volume of 25 cubic feet.

3.14.2 Size 2. - The Size 2 vacuum pumping unit shall be complete with a vertical storage tank approximately 3 feet in diameter by 8 feet high and a minimum volume of 50 cubic feet.

3.14.3 Appurtenances. - The vacuum storage tank shall be complete with legs suitable for vertical mounting. The storage tank shall have a drain valve located on the bottom of the tank, an inlet ball valve, and necessary fittings for complete operation.

3.15 Piping and wiring. - The vacuum pumping unit shall be complete with all necessary piping and wiring for complete operation.

3.15.1 Pipe threads. - Pipe threads shall conform to Specification MIL-P-7105.

3.16 Screw threads. - Screw threads shall conform to Specification MIL-S-7742.

3.17 Cooling system. - The vacuum pump shall be either air or water cooled. When specific requirements exist for a definite type of cooling system, it will be specified (see 6.2).

3.18 Interchangeability. - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.19 Exterior finish. - Exterior surfaces of the vacuum pumping unit shall be finished with two coats of gray gloss enamel, color 512, conforming to MIL-E-7729.

3.20 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

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3.21 Workmanship. - The vacuum pumping unit, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom of blemishes, defects, burrs and sharp edges; accuracy of dimensions, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. - The inspection and testing of the vacuum pumping system shall be classified as follows:

a. Preproduction tests

b. Acceptance tests

4.3 Test conditions. - Unless otherwise specified, all tests required by this specification shall be made at standard atmospheric temperature, and pressure and relative humidity of 80% or less.

4.3.1 Apparatus. - Apparatus used in conjunction with testing prescribed herein shall be, insofar as practicable, of laboratory precision type, and shall be calibrated at intervals properly spaced to insure accuracy.

4.4 Preproduction testing. - Preproduction tests shall consist of all tests described under 4.6.

4.4.1 Preproduction test report. - If the testing is performed by the contractor, he shall prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.5 Acceptance tests. - The acceptance tests shall consist of the following individual test to which each vacuum pumping unit

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shall be subjected:

a. Examination of product . . . . . .See 4.6.1

b. Mechanical inspection . . . . . .See 4.6.2

c. Functional check . . . . . .See 4.6.3

d. Operational check . . . . . .See 4.6.4

e. Vacuum storage tank test . . . . . .See 4.6.5

4.5.1 Rejection and retest. - When one vacuum pumping unit fails to meet this specification, vacuum pumping units still on hand or produced later shall not be accepted until the extent and cause of failure are determined.

4.5.2 Individual tests may continue. - For operational reasons, individual tests may continue pending the investigation of a test failure. Final acceptance of vacuum pumps on hand or produced later shall not be made until it is determined that the vacuum pumps meet all the requirements of this specification.

4.5.3 Defects in vacuum pumping units already accepted. - The investigation of a test failure could indicate that defects may exist in vacuum pumping units already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. - The vacuum pumping unit shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Mechanical inspection. - A mechanical inspection shall be made of all components and parts of the vacuum pump. All pertinent data concerning condition, defects of manufacture, damage in transit, and damage through use prior to test shall be recorded.

4.6.3 Functional check. - All mechanical parts of the vacuum pumping unit shall be carefully checked for free and proper functioning.

4.6.4 Operational check. - The vacuum pumping unit shall be subjected to and shall satisfactorily complete the following tests before acceptance, and shall exhibit no malfunctions, leaks, excessive vibration or other adverse operational characteristics.

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4.6.4.1 Individual proof running. - The vacuum pump shall undergo an operational proof run in excess of 9 hours. The proof run time shall be divided as follows and in the order listed:

a. Three hours of continous operation at rated speed with storage tank inlet valve opened to atmosphere, but throttled to maintain tank pressure at a maximum of 500 mm Hg absolute, and vacuum pumps discharge open to atmosphere.

b. Three hours of continuous operation at rated speed with the vacuum storage tank inlet closed and vacuum pump discharge open to the atmosphere. The vacuum pump control will be set for continuous operation of the pump and vacuum storage tank pressure shall reach and maintain the conditions specified in 3.6.3.

c. Three hours of intermittent operation with vacuum regulator set at one inch intervals from 20 inches of mercury to the ultimate vacuum condition specified in 3.6.3. Each time the vacuum pump discontinues operation, the vacuum storage tank valve shall be cracked and pressures recorded at which the vacuum pump continues operation. The vacuum tank valve will then be closed and vacuum recorded at which the vacuum pump ceases operation. Compliance with 3.12.2 shall be determined.

d. Free air pumping rate. The vacuum pump shall be operated at rated speed until its free air pumping rate, the number of actual cubic feet measured at ambient conditions it will pump through per minute when both inlet and discharge are at atmospheric pressure, has been determined. The free air displacement shall be measured in accordance with the ASME Power Test Code for Displacement Compressors, Blowers and Vacuum Pumps. Compliance with 3.6.1 shall be determined.

e. Using the same test apparatus called for in Paragraph 4.6.4.1d, the pumping unit's Pumping Speed Curve shall be determined over the operating intake pressure range of from atmospheric pressure to blank-off pressure. Three separate tests shall be run and the accumulated data averaged to give one curve. The results of the three specific tests shall also be arranged in a curve and submitted as a part of the test report.

4.6.5 Vacuum storage tank test. - The vacuum storage tank shall be evacuated to the conditions specified in 3.6.3 and isolated from the vacuum pump by shut-off valves. The vacuum storage tank shall be held in this condition for 48 hours. Any increase in storage tank pressure greater than 0.50 inches of mercury shall constitute failure of this test.

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4.6.6 Endurance test. - The vacuum pumping unit having passed the tests specified herein shall be subjected to a 500 hour endurance test divided as follows:

a. The pumping unit shall be operated at rated speed for 250 hours of continuous operation with the vacuum pump discharge open to atmosphere and the vacuum storage tank inlet throttled to maintain a vacuum within the tank of 22 inches Hg. Data for calculation of Pump Capacity in CFM shall be continually recorded.

b. One hundred twenty five hours of continuous operation at rated speed with the vacuum storage tank inlet throttled the vacuum pump discharge open to the atmosphere. The vacuum pump control will be set for continuous operation of the pump and vacuum storage tank pressure shall reach and maintain the conditions specified in 3.6.3. Data for calculation of pump capacity at the highest attained vacuum shall be continually recorded.

c. One hundred twenty five hours of continuous operation with venting and closing off of the vacuum chamber as follows:

(1) 25 minutes continuous operation with vacuum storage tank closed to the atmosphere

(2) 5 minutes continuous operation with vacuum storage tank open to the atmosphere but throttled so as not to exceed an absolute pressure as determined by the contractor. Data for calculation of pumping speed shall be continually recorded.

4.6.7 Inspections, servicings, and adjustments. - Throughout the endurance testing, these inspections, servicings, and adjustments shall be performed during planned stops at 24 hour intervals. Shutdown times shall not exceed one hour. Oil may be changed at 100 hour intervals and added as required.

4.6.8 Observations. - During the progress of all tests, such characteristics as smoothness of operation, vibrations, ease of control, freedom from leaks, ease of handling, and the operation of auxiliary equipment shall be observed to determine whether the vacuum pumping unit complies with all the requirements specified herein.

4.6.9 Performance recheck. - Following the endurance test run and without intervening reconditioning other than normal preventive maintenance and servicing with lubricants, the vacuum pumping unit shall be subjected to a performance recheck. The performance at this time shall not be less than that specified in Section 3 for a new vacuum pump.

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4.7 Preparation for shipment. - Examination of packing and marking for shipment shall be made for conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Prior to preservation, all fuel and oil shall be drained from the vacuum pump. All internal parts of the engine and vacuum pump that may be subject to corrosion shall be thoroughly flushed with a compound conforming to MIL-C-6529.

5.1.1 Level A. - Each vacuum pump shall be preserved in accordance with Method II of MIL-P-116 (see 6.2).

5.1.2 Level C. - Vacuum pumps shall be preserved and packaged in accordance with the manufacturer's commercial practice (see 6.2).

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. - Vacuum pumps preserved and packaged as specified in 5.1 and 5.1.1 shall be packed in exterior shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or MIL-B-10377. Insofar as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B. - Vacuum pumps preserved and packaged as specified in 5.1.1 shall be packed in domestic exterior containers conforming to PPP-B-585 PPP-B-591, PPP-B-601, PPP-B-621, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. - Packages that require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules, or regulations of other common carriers, as applicable to the mode of transportation.

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5.3 Marking for shipment and storage. - In addition to any special markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The vacuum pump covered by this specification is for installation at laboratories at US Air Force activities, and is to be used as a general source of vacuum for testing gyroscopic aircraft instruments, altimeters, speed indicators, and suction gages.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Size of vacuum pumping unit required (see 1.2)

c. Electric motor drive, if other than specified (see 3.10)

d. Cooling system, if definite type of cooling system is required (see 3.17)

e. Level of packaging and packing required (see 5.1.1, 5.1.2 and 5.2)

Reviewer:

MOA (69)

Preparing Activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the spec- ion in procurement of products for ultimate use by the Department of Defense. This sheet is provided for ob- t information on the use of this specification which will insure that suitable products can be procured with a s amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on ex reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

TICATION

ZATION *(Of submitter)* | CITY AND STATE

CT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

AL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

AGENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

MARKS *(Attach any pertinent data which may be of use in improving this specification. If there are addi- onal papers, attach to form and place both in an envelope addressed to preparing activity)*

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